

1. For a function to be analytic at a point, what must hold true?
 - A) Its Laurent series converges at that point
 - B) Its Taylor series converges at that point
 - C) It has singularities at that point
 - D) It has poles at that point
2. If a vector field in cartesian coordinates is $F = i(x^2 - y^2) + j(2xy - z^2) + k(2xz)$, then:
 - A) $\nabla \times F = -4i + 0j + 2k$
 - B) $\nabla \times F = -4i + 2j + 0k$
 - C) $\nabla \times F = 0i - 4j + 2k$
 - D) $\nabla \times F = 0i + 2j - 4k$
3. The eigenvalues of the matrix $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$ are given by:
 - A) 1, 2, 3
 - B) 1, 3, -1
 - C) 1, 0, -1
 - D) 0, 1, 2
4. The residue of $\frac{1}{(1-i)^2}$ at $z = i$ is:
 - A) 1
 - B) 0
 - C) -1
 - D) Undefined
5. The transformation from laboratory to center-of-mass (CM) frame affects which of the following quantities in an elastic collision?
 - A) Magnitude of total momentum
 - B) Relative velocity between particles
 - C) Total energy
 - D) Angle of scattering
6. A Lagrangian with a velocity-dependent potential $V(q, \dot{q})$ leads to:
 - A) Symmetric equations of motion
 - B) Time-reversible trajectories
 - C) Modified Euler-Lagrange equations
 - D) Non-conservative system
7. The rest mass of an electron is m_0 , when it moves with the velocity of $0.6c$ then its mass is:
 - A) $1.6 m_0$
 - B) $1.5 m_0$
 - C) $1.25 m_0$
 - D) $1.8 m_0$
8. A charged particle in a magnetic field has constant speed but curved path. Which force law causes this?
 - A) Newton's third law
 - B) Conservative force
 - C) Lorentz force
 - D) Fictitious force

9. In Hamilton-Jacobi theory, the principal function S satisfies:
- A) $\frac{\partial S}{\partial t} + H = 0$ B) $\frac{\partial S}{\partial t} = L$
- C) $\frac{\partial S}{\partial t} = H$ D) $H = \nabla S \cdot \vec{p}$
10. A particle of charge q and mass m , which is moving in a one-dimensional harmonic potential of frequency ω , is subject to a weak electric field $\mathcal{E}$ in x -direction, what would be the exact expression for the energy?
- A) $\left(n + \frac{1}{2}\right) \hbar \omega + \frac{q^2 \mathcal{E}^2}{2m\omega^2}$ B) $\left(n + \frac{1}{2}\right) \hbar \omega - \frac{q^2 \mathcal{E}^2}{2m\omega^2}$
- C) $\left(n - \frac{1}{2}\right) \hbar \omega + \frac{q^2 \mathcal{E}^2}{2m\omega^2}$ D) $\left(n - \frac{1}{2}\right) \hbar \omega - \frac{q^2 \mathcal{E}^2}{2m\omega^2}$
11. In the high-temperature limit, the Fermi-Dirac and Bose-Einstein distributions reduce to -----distribution.
- A) Maxwell-Boltzmann B) Planck's
- C) Gaussian D) Poisson
12. The entropy in the microcanonical ensemble is defined as:
- A) $S = k_B \ln \Omega$ B) $S = -k_B \sum p_i \ln p_i$
- C) $S = -\frac{\partial F}{\partial T}$ D) $S = k_B T \ln Z$
13. A point charge is placed at the centre of a spherical cavity within a conductor. What is the total induced charge on the inner surface of the cavity?
- A) Zero
- B) Equal and opposite to the point charge
- C) Equal and same sign as the point charge
- D) Twice the point charge
14. A rectangle of cross-sectional area A is placed in a uniform electric field. The normal to the area of the coil make an angle of 90° with the electric field. What will be the electric flux ϕ , through the rectangle?
- A) E B) $\frac{A \cdot E}{\sqrt{2}}$ C) $A \cdot E$ D) Zero
15. The magnetic vector potential due to magnetic dipole is proportional to:
- A) r^{-3} B) r^{-2} C) r^{-1} D) None of these
16. A dielectric slab is inserted halfway into a parallel plate capacitor connected to a constant voltage source. What happens to the force on the slab?
- A) It experiences no net force
- B) It is pulled in due to induced charges
- C) It is pushed out by polarization
- D) It oscillates between plates

17. A molecule in an excited electronic state relaxes to the ground electronic state without changing vibrational level. Which principle is violated?
 A) Heisenberg uncertainty B) Born-Oppenheimer approximation
 C) Frank-Condon principle D) Pauli exclusion principle
18. In NMR spectroscopy, the term 'chemical shift' refers to the:
 A) Resonant frequency of a nucleus
 B) Splitting of NMR signals
 C) Influence of neighboring atoms on NMR signals
 D) Presence of isotopes in the sample
19. The Raman effect involves inelastic scattering of photons due to interaction with --- motion.
 A) Electronic B) Rotational C) Vibrational D) Translational
20. Which configuration of helium will exhibit spectral splitting due to exchange interaction and singlet-triplet splitting?
 A) He^+ ion
 B) Excited state with two electrons in different orbitals
 C) Ground state with paired spins
 D) Ionized helium with one proton and one electron
21. Which of the following would enhance population inversion in a 3-level laser?
 A) Decreasing pump power
 B) Increasing spontaneous emission
 C) Increasing decay rate from upper to intermediate level
 D) Equal population of all levels
22. A stable nucleus with even Z and even N is likely to have:
 A) High spin B) Odd parity
 C) Spin 0 and positive parity D) Negative parity
23. Which of the following is an allowed beta decay based on conservation laws?
 A) $n \rightarrow p + e^+ + \nu_e$ B) $p \rightarrow n + e^+ + \nu_e$
 C) $n \rightarrow p + e^- + \bar{\nu}_e$ D) $p \rightarrow n + \mu^+ + \nu_\mu$
24. The nuclear decay allowed by parity violation in weak interaction:
 A) Alpha decay B) Beta decay
 C) Gamma decay D) Spontaneous fission
25. Identify the pair of physical effects which generally helps distinguish a metal from a semiconductor at low temperatures:
 A) Elastic modulus and bandgap
 B) Phonon dispersion and thermal expansion
 C) Electronic specific heat and electrical conductivity
 D) Fermi level position and lattice vibrations

26. What can be inferred from a material's phonon dispersion relation has an optical branch that flattens near the Brillouin zone boundary?
- A) No atomic motion B) Group velocity is near zero
C) Phase velocity diverges D) Sound speed is infinite
27. Match the following
- | List I | List II |
|-------------------------|-------------------------------------|
| a. Drude model | 1. $S = \frac{-\pi k_B^2 T}{3eE_F}$ |
| b. Wiedemann-Franz law | 2. $R_H = \frac{1}{ne}$ |
| c. Hall effect | 3. $\sigma = \frac{ne^2\tau}{m}$ |
| d. Thermoelectric power | 4. $\frac{K}{\sigma T} = L_0$ |
- A) a-2, b-3, c-4, d-1 B) a-3, b-4, c-1, d-2
C) a-4, b-3, c-1, d-2 D) a-3, b-4, c-2, d-1
28. In band theory, why does a completely filled band **not** contribute to electrical conductivity?
- A) All electrons are at the Fermi level
B) Electron velocity is zero
C) Net electron acceleration is zero due to symmetry
D) Band gap is large
29. Dislocations in crystals primarily contribute to:
- A) Bandgap narrowing B) Phonon scattering
C) Increased symmetry D) Electron acceleration
30. In an RC Coupled amplifier, frequency response is improved with:
- A) More Bias B) Less Gain C) Lower R_1 D) Higher C_C
31. A Class B amplifier is used in push-pull configuration to reduce distortion. This configuration minimize ----- distortion.
- A) Harmonic B) Thermal C) Crossover D) Phase
32. A multistage amplifier has three identical CE stages, each with a gain of 10. The approximate total voltage gain is:
- A) 10 B) 20 C) 100 D) 1000
33. In a Hartley oscillator, increasing the inductance in the tank circuit will result in:
- A) Higher gain B) Lower oscillation frequency
C) Higher frequency drift D) Voltage clipping

34. An ideal operational amplifier has ---- input impedance and ----gain.
 A) Infinite , zero B) Zero, infinite
 C) Infinite, infinite D) Zero, zero
35. An op-amp has a gain-bandwidth product of 1 MHz. What is the maximum gain you can achieve at 10 kHz?
 A) 10 B) 100 C) 1000 D) 10000
36. The logic gate which provides an output '1' only when odd number of its inputs are '1':
 A) OR gate B) NAND gate C) XOR gate D) NOR gate
37. The Stoke's theorem is :
 A) $\iint_s \vec{A} \cdot d\vec{s} = \oint_C \vec{A} \cdot d\vec{r}$
 B) $\oint_C \vec{A} \cdot d\vec{r} = \iint_s (\text{curl } \vec{A}) \cdot d\vec{s}$
 C) $\iint_s \vec{A} \cdot d\vec{s} = \iiint_s (\text{div } \vec{A}) dv$
 D) $\iint_s \vec{A} \cdot d\vec{s} = \iiint_s (\text{grad } \vec{A}) dv$
38. The diagonal elements of which matrix must be real?
 A) Symmetric B) Skew Symmetric
 C) Hermitian D) Skew Hermitian
39. The complete solution of the differential equation:
 $\frac{d^2y}{dx^2} - 7 \frac{dy}{dx} + 6y = e^{2x}$ is:
 A) $C_1 e^x + C_2 e^{3x} - e^{2x}$ B) $C_1 e^x + C_2 e^{6x} - e^{2x}$
 C) $C_1 e^x + C_2 e^{6x} - \frac{1}{4} e^{2x}$ D) $C_1 e^x + C_2 e^{3x} - \frac{1}{4} e^{2x}$
40. Which of the following functions of complex variable $z = x + iy$ is analytic?
 A) $|z|$ B) $\text{Re } Z$ C) $\log Z$ D) z^{-1}
41. The locus represented by $|z - 3| + |z + 3| = 10$ is:
 A) Hyperbola B) Parabola C) Circle D) Ellipse
42. The value of the integral $I = \int_0^{2\pi} \frac{\cos 2\theta d\theta}{(5+4 \cos \theta)}$ is:
 A) $\frac{\pi}{6}$ B) $\frac{\pi}{4}$ C) $\frac{\pi}{2}$ D) π

43. The Fourier series for $f(x) = \sin^2 x$ defined over the range $-\pi \leq x \leq \pi$ is:
 A) $\frac{1}{2} - \frac{\cos 2x}{2}$ B) $1 + \cos 2x$ C) $\frac{1}{2} - \frac{\cos x}{2}$ D) $\frac{1}{2} + \frac{\cos 2x}{2}$
44. If $f(t) = e^{at}$, its Laplace transform (for $s > a$) is given by:
 A) $\frac{a}{s^2} + (s - a)$ B) $\frac{\sqrt{\pi}}{2(s-a)}$ C) $\frac{1}{(s-a)}$ D) Does not exist
45. $x\sqrt{1+y} + y\sqrt{1+x} = 0$, then $\frac{dy}{dx}$
 A) $(1+x)$ B) $(1+x)^{-2}$ C) $-(1+x)^{-1}$ D) $-(1+x)^{-2}$
46. The Rodrigue's representation of Laguerre's polynomial $L_n(x)$ is:
 A) $\frac{n!}{e^x} \frac{d^n}{dx^n} (x^{-n} e^x)$ B) $\frac{e^x}{n!x} \frac{d^n}{dx^n} (x^{-n} e^x)$
 C) $\frac{n!}{e^x} \frac{d^n}{dx^n} (x^n e^{-x})$ D) $\frac{e^x}{n!x} \frac{d^n}{dx^n} (x^n e^{-x})$
47. For Dirac Delta function, $\int_{-1}^1 x^2 \delta(3x+1) dx = \dots$.
 A) $\frac{4}{27}$ B) $\frac{1}{27}$ C) $\frac{4}{9}$ D) $\frac{1}{3}$
48. A particle is moving in a central force field given by $\vec{F} = -kr^n \hat{r}$ where k is a constant, r is the distance of the particle from the origin and $\hat{r}$ is the unit vector in the direction of r . For which values of n are closed and stable orbits possible?
 A) $n = 1$ and $n = 2$ B) $n = 1$ and $n = -2$
 C) $n = 1$ and $n = -1$ D) $n = 2$ and $n = -2$
49. A satellite is revolving around the earth with a speed of 7.6 km/s what is the height of the satellite from the earth surface, consider the mass of the earth = 6×10^{24} kg and radius of the earth = 6400 km.
 A) 500 km B) 550 km C) 600 km D) 650 km
50. In Rutherford scattering cross section, differential scattering cross section is proportional to:
 A) e B) e^2 C) e^3 D) e^4
51. The value of m and n for which the transformations are $Q = q^m \cos np$; $P = q^m \sin np$ represents canonical transformations are:
 A) $m = 2, n = \frac{1}{2}$ B) $m = \frac{1}{2}, n = 2$
 C) $m = 1, n = 2$ D) $m = 2, n = 1$

52. For a charged particle in an electromagnetic field, the Hamiltonian $\vec{H}$ is represented as:
- A) $\frac{1}{2}m\left(\frac{p}{m} + \frac{q}{mc}\vec{A}\right)^2 + q\phi$ B) $\frac{1}{2}m\left(\frac{p}{m} + \frac{q}{mc}\vec{A}\right)^2 - q\phi$
 C) $\frac{1}{2}m\left(\frac{p}{m} - \frac{q}{mc}\vec{A}\right)^2 + q\phi$ D) $\frac{1}{2}m\left(\frac{p}{m} - \frac{q}{mc}\vec{A}\right)^2 - q\phi$
53. A proton and an alpha particle are accelerated in a field of the same potential difference. Then the ratio of the de Broglie wavelengths associated with the moving material particles is:
- A) $2\sqrt{2}:1$ B) $1:2$ C) $4:1$ D) $2:1$
54. Number of possible modes of vibration perpendicular to the axis in linear symmetric triatomic molecules :
- A) One B) Two C) Three D) Four
55. The Poisson bracket $[X, Y]$ of $X = q_1^2 + q_2^2$ and $Y = 2P_1 + P_2$ is:
- A) 0 B) $(q_1^2 + q_2^2)P_1$
 C) $4q_1 + 2q_2$ D) $3(q_1^2 + q_2^2)$
56. The product of generalized co-ordinate and its conjugate momentum has the dimensions of:
- A) Linear momentum B) Angular momentum
 C) Force D) Energy
57. An astronaut at rest has a heart rate of 65 beats/min, when the astronaut's spaceship moves at a speed of $0.6c$, her heart rate as measured by an earth observer will be ---- beats/min.
- A) 26 B) 104 C) 74 D) 52
58. If elements with principal quantum number $n > 4$ were not allowed in nature, the number of possible elements would be:
- A) 60 B) 64 C) 32 D) 4
59. The duration of radar pulse is $10^{-6}s$. The uncertainty in its energy would be:
- A) 0 B) $1.05 \times 10^{-21}J$
 C) $1.05 \times 10^{-28}J$ D) $1.05 \times 10^{-35}J$
60. The wave function in the ground state of hydrogen atom is given as $\psi = Ae^{-r/a}$, where r measures distance from nucleus and a is constant, the value of A is:
- A) $\frac{1}{\sqrt{\pi a}}$ B) $\frac{1}{\sqrt{\pi a^3}}$ C) $\frac{1}{\sqrt{\pi a^2}}$ D) $\frac{1}{\sqrt{\pi a^5}}$

61. The degree of degeneracy for three-dimensional isotropic harmonic oscillator is:
 A) n^2 B) $2n+1$
 C) $\frac{1}{2}(2n+1)(2n+2)$ D) $\frac{1}{2}(n+1)(n+2)$
62. In Born approximation, the scattering amplitude is:-
 A) $\frac{-1}{k} \int_0^\infty r \sin^2 kr V(r) dr$ with $k = 2k \sin \frac{\theta}{2}$
 B) $\frac{-1}{k} \int_0^\infty \sin kr V(r) dr$ with $k = 2k \sin \frac{\theta}{2}$
 C) $\frac{-1}{k} \int_0^\infty r \sin kr V(r) dr$ with $k = 2k \sin \frac{\theta}{2}$
 D) $\frac{-1}{k} \int_0^\infty r^2 \sin kr V(r) dr$ with $k = 2k \sin \frac{\theta}{2}$
63. The phase velocity of ocean wave is $\sqrt{\frac{g\lambda}{2\pi}}$, then the group velocity of ocean wave is:
 A) $\frac{V_p}{2}$ B) V_p C) 0 D) $\frac{V_p}{4}$
64. If $\bar{J}_1$ and $\bar{J}_2$ are angular momenta of two non-interacting systems and $\bar{J}$ represents the sum of $\bar{J}_1$ and $\bar{J}_2$ then ---- commute with -----.
 A) J_1^2 and J_2^2 , J^2 only B) J_1^2 and J_2^2 , J_z only
 C) J_1^2 and J_2^2 , both J^2 and J_z D) J_{1z} and J_{2z} , J^2 only
65. The expression for optical theorem is:
 A) $I_m f(0) = \frac{k}{4\pi}$ B) $I_m f(0) = \frac{k}{4\pi} \sigma_T$
 C) $Re f(0) = 2\hbar i$ D) $I_m f(0) = k\pi \sigma_T$
66. The normalization constant, N , for $\psi = Nre^{-ar}$ for $0 < r < \infty$ is:
 A) $2\sqrt{a^3}$ B) $2a^2$ C) $3\sqrt{a^3}$ D) $3a^2$
67. The energy functional from a trivial wave function is $E(\alpha) = (\alpha^2 - 3\alpha) / 6$. The variationally optimized energy is:
 A) $-\frac{1}{2}$ B) $-\frac{3}{8}$ C) $\frac{3}{2}$ D) $\frac{3}{8}$
68. $Axe^{-x^2/2}$ is an eigen function of $\hat{H}$, what is the associated eigen value?
 A) 1 B) 2 C) 3 D) 4

69. The first order correction to the ground state energy of a particle in a well with a variation in the potential of the form $V = -\epsilon \sin\left(\frac{\pi x}{L}\right)$ is:
- A) $-\frac{\epsilon}{3\pi}$ B) $-\frac{2\epsilon}{3\pi}$ C) $-\frac{4\epsilon}{3\pi}$ D) $-\frac{8\epsilon}{3\pi}$
70. The Maxwell's relation which leads to Clausius – Clapeyron equation:
- A) $\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial P}{\partial V}\right)_T$ B) $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_T$
 C) $\left(\frac{\partial T}{\partial P}\right)_S = \left(\frac{\partial V}{\partial S}\right)_P$ D) $\left(\frac{\partial V}{\partial T}\right)_P = -\left(\frac{\partial S}{\partial P}\right)_T$
71. A Certain system of non-interacting particles has the single particle partition function $f = A \frac{V}{T^m}$, where A is constant. The average energy per particle will be:
- A) $mk_B T$ B) $-mk_B T$ C) $\frac{k_B T}{m}$ D) $-\frac{k_B T}{m}$
72. Which of the following relations between total energy E and the microcanonical partition function Z is true?
- A) $E = -N kT \log Z$ B) $E = N kT \frac{\partial}{\partial V}(\log Z)$
 C) $E = N kT^2 \left[\frac{\partial(\log Z)}{\partial T}\right]_V$ D) $E = -N \frac{\partial}{\partial T} \log Z$
73. Four distinguishable particles fill up energy levels $0, \epsilon, 2\epsilon$. The number of available microstates for the total energy 4ϵ is:
- A) 20 B) 24 C) 11 D) 19
74. If a polished metal plate which has a rough and black spot is heated to 1400K and brought immediately in a dark room, the spot:
- A) Will appear brighter than the plate.
 B) Will appear darker than the plate.
 C) And plate will appear equally bright.
 D) And the plate will not be visible in the dark room.
75. If the distribution of energy radiated by a black body, Planck's law reduces to the Rayleigh Jeans law for :
- A) Shorter wavelengths B) Longer wavelengths
 C) Both (A) and (B) D) No relation between the two
76. Two charges q_1 and q_2 are placed on the x axis with q_1 at $x = a$ and q_2 at $x = 3a$, A third charge Q is placed at the origin. The net force on charge Q is zero if q_2 is equal to:
- A) $3q_1$ B) $-3q_1$ C) $9q_1$ D) $-9q_1$

77. The electromagnetic field equation in terms of electromagnetic potentials (A&B) is represented by an equation :
- A) $\nabla^2 \phi + \frac{\partial}{\partial t}(\text{div } \bar{A}) = -\frac{\rho}{\epsilon}$ B) $\nabla^2 \bar{A} + \frac{\partial}{\partial t}(\text{div } \phi) = -\frac{\rho}{\epsilon}$
- C) $\nabla^2 \phi + \frac{\partial}{\partial t}(\text{div } \bar{A}) = \frac{\rho}{\epsilon}$ D) $\nabla^2 \bar{A} + \frac{\partial}{\partial t}(\text{div } \phi) = \frac{\rho}{\epsilon}$
78. For good conductors, skin depth varies inversely with ---- power of frequency.
- A) One B) Two C) Half D) Three
79. An electromagnetic wave is to pass through an interface separating two media having dielectric constants ϵ_1 and ϵ_2 respectively. If $\epsilon_1 = 4\epsilon_2$, the wave will be totally reflected if the angle of incidence is:
- A) 0° B) 30° C) 45° D) 60°
80. In a dispersive medium, the group velocity is:
- A) Less than the phase velocity only
- B) Equal to the phase velocity only
- C) More than the phase velocity, depending on the nature of the dispersive medium
- D) More than the phase velocity
81. The energy stored in a magnetic field of lengths 1m and of cross section A square meter is:
- A) $E = \frac{1}{2} \times A \frac{N^2}{\mu l} \times I^2 J$ B) $E = \frac{1}{2} \times \frac{\mu A N^2}{l} \times I^2 J$
- C) $E = \frac{1}{2} \times \mu l \frac{N^2}{A} \times I^2 J$ D) $E = \frac{1}{2} \times \mu A \frac{N^2}{l} \times I^2 J$
82. If the electrostatic potential were given by $\phi = \phi_0(x^2 + y^2 + z^2)$ where ϕ_0 is constant, then the charge density giving rise to the above potential would be:
- A) 0 B) $-2\phi_0\epsilon_0$ C) $-6\phi_0\epsilon_0$ D) $-\frac{6\phi_0}{\epsilon_0}$
83. If there is a dispersive medium whose refractive index depends on the wavelength as given by $(\lambda) = n_0 + \frac{a}{\lambda^2} - \frac{b}{\lambda^4}$. The value of λ at which the group and phase velocities would be same, is:
- A) $\sqrt{\frac{b}{2a}}$ B) $\sqrt{\frac{2b}{a}}$ C) $\sqrt{\frac{3b}{a}}$ D) $\sqrt{\frac{b}{3a}}$
84. The ratio of frequencies of the first line of the Lyman series and the first line of Balmer series is:
- A) $\frac{4}{27}$ B) $\frac{8}{27}$ C) $\frac{27}{5}$ D) $\frac{27}{8}$

85. A proton, a neutron, an electron and an α particle have the same energy. If $\lambda_p, \lambda_n, \lambda_e$ and λ_α are their de Broglie's wavelengths, then choose the correct relation from the following:
- A) $\lambda_e < \lambda_p = \lambda_n > \lambda_\alpha$ B) $\lambda_\alpha < \lambda_n < \lambda_p < \lambda_e$
 C) $\lambda_e = \lambda_p = \lambda_n = \lambda_\alpha$ D) $\lambda_\alpha > \lambda_n > \lambda_p > \lambda_e$
86. The magnetic moment associated with the first orbit in hydrogen atom is given by:
- A) $\frac{h}{4\pi m e}$ B) $\frac{4\pi m}{eh}$ C) $\frac{eh}{4\pi m}$ D) $\frac{ehm}{4\pi}$
87. Atoms with $\frac{1}{2}$ nuclear spin cannot have :
- A) Fine structure B) Electric dipole interaction
 C) Hyperfine structure D) None of these
88. The degeneracy of the spectral term 3F is:
- A) 15 B) 9 C) 7 D) 21
89. The Lande g-factor for the 3P_1 level of an atom is:
- A) $\frac{7}{2}$ B) $\frac{5}{2}$ C) $\frac{3}{2}$ D) $\frac{1}{2}$
90. At higher magnetic fields, the splitting of spectral lines is disturbed. This effect is called ---- effect.
- A) Stark B) Inverse Zeeman
 C) Anomalous Zeeman D) Paschen-back
91. Which one of the following is accompanied by the characteristic X-Ray emission?
- A) Positron emission B) Electron emission
 C) Helium nucleus emission D) K-electron capture
92. The molecule that will **not** absorb in the microwave region, but will absorb in the infrared is:
- A) HCl B) C_2H_2 C) H_2O D) N_2
93. The energy change in an electronic transition is of the order of:
- A) 5 – 10 eV B) 10^{-2} eV C) 0.1 eV D) 1 keV
94. The radius of a spherical nucleus measured by electron scattering is 3.6 fm. What is the mass number of the nucleus most likely to be?
- A) 27 B) 40 C) 56 D) 120
95. The asymmetry terms in the Weizsacker semi-empirical mass formulae is because of:
- A) Non-zero spin of nucleus
 B) Unequal number of protons and neutrons inside the nucleus
 C) Non-spherical shape of the nucleus
 D) Odd number of protons inside the nucleus

96. Which of the following decay processes is allowed?
 A) $k^0 \rightarrow \mu^+ + \mu^-$ B) $\mu^- \rightarrow e^- + \gamma$
 C) $n \rightarrow p + \pi^-$ D) $n \rightarrow \pi^+ + \pi^-$
97. The tensor component of the nuclear force may be inferred from the fact that deuteron nucleus 2_1H :
 A) Has only one bound state with total spin $s = 1$
 B) Has a non-zero electric quadrupole moment in its ground state
 C) Is stable while triton 3_1H is unstable
 D) Is the only two nucleon bound state
98. In Yukawa Meson theory of nuclear force, the potential for the nuclear force between neutron and proton is assumed as:
 A) $-g^2 \frac{e^{\lambda r}}{r^2}$ B) $-g^2 \frac{e^{-\lambda r}}{r}$ C) $-g^2 \frac{e^{-\lambda r^2}}{r}$ D) $-g^2 \frac{e^{\lambda r^2}}{r^3}$
99. The mean life of the excited state of Cd is 8×10^{-8} sec, the half-width of the excited state is :
 A) 10^{-2} eV B) 10^{-4} eV C) 10^{-6} eV D) 10^{-8} eV
100. The quarks are supposed to exist in ---- number of flavours.
 A) Two B) Four C) Six D) Sixteen
101. The interaction potential between two quarks, separated by a distance r inside a nucleon, can be described by:
 A) $ae^{-\beta r}$ B) $\frac{a}{r}$ C) $\frac{a}{r} + br$ D) $-\frac{a}{r} + br$
102. According to quark model, it is possible to build all the hadrons using ---- quarks and ---- antiquarks.
 A) 3, 2 B) 3, 3 C) 2, 2 D) 2, 3
103. The unit cell of a certain type of crystal is defined by $a \neq b \neq c$ and the angles are mutually perpendicular. The crystal structure is:
 A) Monoclinic B) Tetragonal
 C) Orthorhombic D) Triclinic
104. Atomic radius of BCC structure is:
 A) $\frac{4a}{\sqrt{3}}$ B) $\frac{\sqrt{3}a}{4}$ C) $\frac{a}{2\sqrt{2}}$ D) $\frac{a}{2}$
105. If the Fermi energy of silver at '0' K is 5 eV, the mean energy of electron in silver at '0' K is:
 A) 1.5 eV B) 3 eV C) 6 eV D) 12 eV

106. Electronic contribution to the specific heat of a metal at low temperature is:
 A) Zero B) An exponential function of T
 C) A linear function of T D) None of these
107. The London penetration depth is given by:
 A) $\lambda = \left[\frac{m}{n_s \mu_0 e^2} \right]^{\frac{1}{2}}$ B) $\lambda = \left[\frac{n_s}{m \mu_0 e^2} \right]^{\frac{1}{2}}$
 C) $\lambda = [n_s m \mu_0 e^2]^{\frac{1}{2}}$ D) None of these
108. If $\vec{g}$ is a reciprocal lattice vector, the Bragg's law can be written as:
 A) $\vec{k} + \vec{g} = 0$ B) $2\vec{k} \cdot \vec{g} + g^2 = 0$
 C) $2\vec{k} \cdot \vec{g} + k^2 = 0$ D) $\vec{k} \cdot \vec{g} = 0$
109. In ---- materials, the susceptibility will decrease with increase in temperature, and they have relatively small susceptibility at all temperatures.
 A) Ferromagnetic B) Ferrimagnetic
 C) Superparamagnetic D) Antiferromagnetic
110. Frenkel defect is due to the existence of:
 A) Extra electrons in the crystal
 B) Extra holes in valence band
 C) Lattice distortions in crystal
 D) None of these
111. The reflecting plane of LiF crystal has an interplanar distance of 4Å. Calculate the wavelength of second order diffraction line which has a glancing angle of 30°:
 A) 1Å B) 2 Å C) 3 Å D) 4 Å
112. Which among the following is a current controlled device?
 A) JFET B) IGBT C) BJT D) MOSFET
113. The diode used in ultra-high speed switching electronic circuits is ----- diode.
 A) Zener B) Varactor C) Tunnel D) Schottky
114. A JFET always operates with:
 A) The gate-to-source pn junction reverse biased
 B) The gate-to-source pn junction forward biased
 C) The gate connected to the source
 D) The drain connected to the ground
115. In direct-coupled amplifier, the gain falls at high frequencies mainly due to:
 A) Filtering issues B) Stray capacitances
 C) Power supply D) Coupling capacitors

116. The oscillators which can be used, where high stability of frequency is required:
- | | |
|-----------------------|---------------------------|
| A) Crystal oscillator | B) Wien bridge oscillator |
| C) Hartley oscillator | D) Colpitts oscillator |
117. CMRR for an op-amp should be:
- | | |
|-------------------------|-------------------------|
| A) As small as possible | B) Close to zero |
| C) Close to unity | D) As large as possible |
118. In 8085 microprocessors, the first machine cycle of every instruction is:
- | | |
|-----------------------|-----------------------|
| A) Memory Read Cycle | B) Memory Write Cycle |
| C) Opcode Fetch Cycle | D) I/O Read Cycle |
119. The size of internal data memory in an 8051 microcontroller is ----- bytes.
- | | | | |
|---------|---------|--------|--------|
| A) 2048 | B) 1024 | C) 256 | D) 128 |
|---------|---------|--------|--------|
120. If the registers have both shifts and parallel load capabilities, they are referred as---- Registers.
- | | |
|--------------------|-------------------------|
| A) Universal Shift | B) Unidirectional Shift |
| C) Sequential | D) Bidirectional Shift |
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